Category Theory for Mathematical Physics — Exercise Sheet 0

Diagnostic: what the course assumes

Variant: Hints and answers

This optional sheet tests the prerequisites; each exercise names the sibling lectures that teach them and the lectures of this course that use them. Vector spaces are finite-dimensional over $\mathbb{C}$, and in $g \circ f$ the map f acts first. For V with basis $e_1, \dots, e_n$ ($n = \dim V$), the dual basis of V^* is $e^1, \dots, e^n$, $e^i(e_j) = \delta_{ij}$; $m = \dim W$. P is the flip $v \otimes w \mapsto w \otimes v$, and the partial trace $\mathrm{tr}_2 \colon \mathrm{End}(V \otimes W) \to \mathrm{End}(V)$ is $\mathrm{tr}_2(M) = \sum_k (\mathrm{id} \otimes e^k)\, M\, (\mathrm{id} \otimes e_k)$ for a basis e_k of W. Groups are multiplicative, with identity 1. The singlet is written $|\psi^-\rangle$, and the antisymmetric tensor is $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1$, as in Lecture 4.

Exercise 1 (Duals and the trace). (*Linear Algebra*, Lectures 11–12; used from Lecture 4.) Let $\mathrm{ev} \colon V^* \otimes V \to \mathbb{C}$, $\varphi \otimes v \mapsto \varphi(v)$, and $\mathrm{coev} \colon \mathbb{C} \to V \otimes V^*$, $1 \mapsto \sum_i e_i \otimes e^i$.

(i) Show that coev does not depend on the basis.

(ii) Show that $(\mathrm{id}_V \otimes \mathrm{ev}) \circ (\mathrm{coev} \otimes \mathrm{id}_V) = \mathrm{id}_V$ and $(\mathrm{ev} \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{V^*} \otimes \mathrm{coev}) = \mathrm{id}_{V^*}$.

(iii) Show that $\mathrm{ev} \circ P \circ (f \otimes \mathrm{id}_{V^*}) \circ \mathrm{coev} = \mathrm{tr}\, f$ for $f \colon V \to V$; in particular $\mathrm{ev} \circ P \circ \mathrm{coev} = \dim V$.

(iv) Show that $\varphi \otimes w \mapsto \big(v \mapsto \varphi(v)\, w\big)$ is an isomorphism $V^* \otimes W \to \mathrm{Hom}(V, W)$.

Hint. (i) A new basis $f_k = \sum_i B_{ik} e_i$ has dual basis $f^k = \sum_j (B^{-1})_{kj} e^j$. (iv) Compare dimensions, or write the inverse.

Answer. (i) $\sum_k f_k \otimes f^k = \sum_{i,j} (BB^{-1})_{ij}\, e_i \otimes e^j = \sum_i e_i \otimes e^i$. (ii) The first sends e_j to $\sum_i e_i\, e^i(e_j) = e_j$, the second φ to $\sum_i \varphi(e_i) e^i = \varphi$. (iii) $\sum_i e^i(f e_i) = \mathrm{tr}\, f$. (iv) It is onto, with inverse $g \mapsto \sum_i e^i \otimes g(e_i)$, and both spaces have dimension nm.

Exercise 2 (Tensor products of maps). (*Linear Algebra*, Lecture 12; *Quantum Mechanics I*, Lectures 10–11; used from Lecture 3.)

(i) Show that $(f \otimes g) \circ (f' \otimes g') = ff' \otimes gg'$ and $P \circ (f \otimes g) \circ P = g \otimes f$ for $f \colon V \to V$ and $g \colon W \to W$, where the right P is the flip $W \otimes V \to V \otimes W$ and the left one $V \otimes W \to W \otimes V$.

(ii) On $\mathbb{C}^2 \otimes \mathbb{C}^2$, find the eigenvalues of P with their eigenvectors.

(iii) Show that P on $\mathbb{C}^2 \otimes \mathbb{C}^2$ is not of the form $f \otimes g$.

(iv) Show that $\mathrm{tr}_2(A \otimes B) = (\mathrm{tr}\, B)\, A$.

Hint. (i) Check on $v \otimes w$. (iii) Apply a candidate $f \otimes g$ to $e_1 \otimes e_1$ and to $e_1 \otimes e_2$.

Answer. (i) Both sides agree on every $v \otimes w$. (ii) $+1$ on $e_1 \otimes e_1$, $e_2 \otimes e_2$, $e_1 \otimes e_2 + e_2 \otimes e_1$; -1 on $e_1 \otimes e_2 - e_2 \otimes e_1$. (iii) $f \otimes g$ fixing $e_1 \otimes e_1$ maps $e_1 \otimes e_2$ into $e_1 \otimes \mathbb{C}^2$, which does not contain $e_2 \otimes e_1$. (iv) tr_2 contracts the second factor.

Exercise 3 (Characters of S_3). (*Symmetries in Physics*, Lectures 3–5 and 8; used from Lecture 2.) Let std be the permutation representation of S_3 on $\mathbb{C}^3$ minus its trivial summand.

(i) Write the characters of **1**, sgn and std on the three conjugacy classes, and check that they are orthonormal.

(ii) Decompose std $\otimes$ std.

(iii) Find $\dim \mathrm{Hom}_{S_3}(\mathrm{std} \otimes \mathrm{std}, \mathrm{std})$, and show that every S_3-map std $\to$ std is a scalar.

Hint. (i) The character of a permutation representation counts fixed points. (iii) Schur's lemma.

Answer. (i) On 1, transpositions, 3-cycles: $(1,1,1)$, $(1,-1,1)$, $(2,0,-1)$. (ii) $\chi^2_{\text{std}} = (4,0,1)$: $\mathbf{1} \oplus \text{sgn} \oplus \text{std}$. (iii) 1; std is irreducible, so Schur gives scalars.

Exercise 4 (Spin and Clebsch–Gordan). (*Symmetries in Physics*, Lectures 12–14; *Quantum Mechanics I*, Lecture 13; used from Lecture 2, and in Lectures 4 and 9.) Let $V_{1/2} = \mathbb{C}^2$ be the defining representation of $\mathrm{SU}(2)$, with basis $e_1 = \uparrow$, $e_2 = \downarrow$, and V_j the spin-j representation.

(i) Show that $V_{1/2} \otimes V_{1/2} \cong V_0 \oplus V_1$, with V_0 spanned by the singlet $(e_1 \otimes e_2 - e_2 \otimes e_1)/\sqrt{2}$, and that the singlet is fixed by $U \otimes U$ for every $U \in \mathrm{SU}(2)$.

(ii) With the characters $\chi_j(t) = \sin\big((2j+1)t/2\big)/\sin(t/2)$ on the rotation by t (extended continuously where the denominator vanishes), show that $\chi_{1/2}\chi_j = \chi_{j-1/2} + \chi_{j+1/2}$ for $j \geq \frac{1}{2}$, and deduce $V_{1/2} \otimes V_j \cong V_{j-1/2} \oplus V_{j+1/2}$.

(iii) Decompose $V_1 \otimes V_1$ and check the dimensions.

Hint. (i) Compute $(U \otimes U)(e_1 \otimes e_2 - e_2 \otimes e_1)$ in terms of the entries of U. (ii) Write $\chi_{1/2}$ in closed form, then use a product-to-sum formula.

Answer. (i) $(U \otimes U)$ multiplies the singlet by $\det U = 1$; the symmetric tensors are spin 1. (ii) The identity is the sine addition; characters determine representations. (iii) $V_0 \oplus V_1 \oplus V_2$: $9 = 1 + 3 + 5$.

Exercise 5 (Two qubits). (*Quantum Mechanics I*, Lectures 10–11; used from Lecture 3.) Let $|\psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ in $\mathbb{C}^2 \otimes \mathbb{C}^2$, with $|0\rangle = e_1$ and $|1\rangle = e_2$.

(i) Show that $|\psi^-\rangle$ is not a product $|a\rangle \otimes |b\rangle$.

(ii) Compute the reduced state $\rho_1 = \mathrm{tr}_2 |\psi^-\rangle\langle\psi^-|$, and compare with tr_2 of a product state.

(iii) How does P act on $|\psi^-\rangle$, and on $(|01\rangle + |10\rangle)/\sqrt{2}$?

Hint. (i) Write a vector $\sum_{ij} c_{ij} |i\rangle |j\rangle$ as the matrix (c_{ij}); a product has rank 1.

Answer. (i) Its matrix $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has rank 2. (ii) $\rho_1 = \frac{1}{2} \cdot 1$, maximally mixed; for $|a\rangle \otimes |b\rangle$ it is the pure $|a\rangle\langle a|$. (iii) -1; $+1$.

Exercise 6 ⋆. *An invariant cup with loop* -2 (a preview of Lecture 4). On $V = \mathbb{C}^2$ let $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1 \in V \otimes V$, and let $\Omega^\dagger : V \otimes V \to \mathbb{C}$ be the linear map with $\Omega^\dagger(e_1 \otimes e_2) = 1$, $\Omega^\dagger(e_2 \otimes e_1) = -1$, $\Omega^\dagger(e_i \otimes e_i) = 0$.

(i) Show that Ω and $\Omega^\dagger$ are invariant under $\mathrm{SU}(2)$: $(U \otimes U)\Omega = \Omega$ and $\Omega^\dagger \circ (U \otimes U) = \Omega^\dagger$.

(ii) Compute $(\mathrm{id} \otimes \Omega^\dagger) \circ (\Omega \otimes \mathrm{id})$ and $(\Omega^\dagger \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \Omega)$ as maps $V \to V$.

(iii) Show that if an invariant cup $\lambda\Omega$ and an invariant cap $\mu\Omega^\dagger$ satisfy both snake identities of Exercise 1, then the loop $(\mu\Omega^\dagger)(\lambda\Omega)$ is -2. Why can no invariant cup and cap that satisfy both snakes make the loop $+2 = \dim V$, as in Exercise 1?

Hint. (i) Both are determinants. (iii) The invariant vectors in $V \otimes V$ are the multiples of Ω (Exercise 4).

Answer. (i) $(U \otimes U)\Omega = \det U\, \Omega$, $\Omega^\dagger(Ux \otimes Uy) = \det U\, \Omega^\dagger(x \otimes y)$. (ii) Both are $-\mathrm{id}$. (iii) The snakes force $\lambda\mu = -1$, and $\Omega^\dagger(\Omega) = 2$; the cup coev $= \sum_i e_i \otimes e^i$ of Exercise 1 lives in $V \otimes V^*$, and identifying V^* with V invariantly costs this sign.